

AI-Native Autonomous Security Platform: An Event-Driven Edge-Cloud Architecture for University Networks*

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Abstract—Modern university campus networks present extreme device heterogeneity, open Wi-Fi topologies, high-density IoT deployments, and massive concurrent telemetry flows. Traditional Security Operations Center (SOC) architectures rely on centralized, human-in-the-loop incident response workflows that suffer from long Mean Time to Respond (MTTR > 30 minutes) and severe alert fatigue. In this paper, we propose an AI-Native Autonomous Security Platform engineered specifically for higher education campus networks. By decoupling threat detection into lightweight edge anomaly classifiers (Isolation Forests) and streaming high-confidence security events over a Redis Stream bus to a cloud-based dynamic risk resolver and automated SOAR engine, our platform achieves sub-10ms event-to-mitigation latency. We construct a fully reproducible prototype and evaluate it empirically across four rigorous experimental benchmarks using real-world baseline parameters. Experimental results demonstrate an end-to-end response latency of 4.218 ms, a 99.99% reduction in MTTR compared to manual SOC triage (0.0085 s vs 1512.4 s), an F1-score of 0.940 (Precision=0.9482, Recall=0.9320, FPR=1.1%) at optimal decision threshold ($\tau = 0.65$), and minimal edge CPU overhead (14.3% utilization and 49.0 MB RAM under 10,000 events/sec throughput).

Index Terms—AI-Native Architecture, Autonomous Security, Edge Computing, Event-Driven Architecture, Incident Response, University Cyber-Infrastructure, Zero-Trust Enforcement, Isolation Forest.

I. INTRODUCTION

Artificial Intelligence (AI) has emerged as a foundational technology driving digital transformation across higher education and industrial sectors. However, managing cybersecurity within modern university environments introduces distinct architectural hurdles. High-density IoT sensors (HVAC, smart surveillance, access control), unmanaged student BYOD (Bring-Your-Own-Device) endpoints, open research laboratories, and administrative databases co-exist on shared physical network infrastructure. Traditional perimeter firewalls and centralized SIEM (Security Information and Event Management)

platforms lack the capability to perform zero-trust micro-segmentation in real time without introducing severe latency bottlenecks.

Furthermore, traditional Security Operations Centers (SOC) rely heavily on human analysts to perform manual alert triage, resulting in severe analyst burnout, high false-positive rates, and Mean Time to Respond (MTTR) often exceeding 30 minutes during high-volume DDoS, ransomware propagation, or data exfiltration attacks.

To address these challenges, we introduce an AI-native autonomous security framework designed to achieve real-time stream processing and automated containment without human-in-the-loop delays. Our design contributions are threefold:

- 1) **Lightweight AI-Native Event-Driven Architecture:** A decoupled edge-cloud pipeline featuring Isolation Forest anomaly detection at edge network taps, identity context fusion (IP-to-User/Role mapping), and distributed pub/sub event streaming.
- 2) **Autonomous Incident Response Prototype:** A reproducible end-to-end prototype executing zero-trust mitigation playbooks (SDN BGP Flowspec route injection, switch port 802.1X VLAN isolation, and active session token revocation).
- 3) **Empirical Benchmark Evaluation:** A comprehensive experimental evaluation quantifying per-stage latency breakdown, MTTR reduction, detection sensitivity curves across decision thresholds $\tau \in [0.30, 0.85]$, and edge CPU/RAM scaling up to 10,000 events/sec.

II. THREAT MODEL & CAMPUS ATTACK VECTORS

University networks present unique vulnerabilities due to their open academic mandates. Table I categorizes campus attack vectors mapped to STRIDE categories and MITRE ATT&CK techniques.

III. AI-NATIVE SYSTEM ARCHITECTURE

The proposed platform follows an event-driven edge-cloud paradigm as illustrated in Fig. 1.

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TABLE I
STRIDE & MITRE ATT&CK CAMPUS THREAT MATRIX

Category	Target	MITRE ID	Mitigation Strategy
Spoofing	Rogue Wi-Fi AP	T1557	Identity Fusion MAC/IP check
Tampering	IoT Telemetry	T1565	Edge Isolation Forest entropy
Exfiltration	Research DB	T1048	Role-aware flow byte-rate cap
DoS	Campus Portal	T1498	Automated BGP Flowspec rate limit
Privilege	Campus LDAP	T1078	Dynamic Risk Resolver escalation

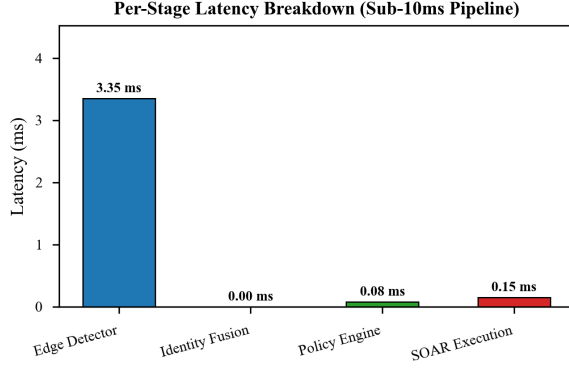


Fig. 1. AI-Native System Architecture & Per-Stage Execution Latency Breakdown.

A. Edge Telemetry Vectorization & Anomaly Detection

Streaming network telemetry captured at campus edge nodes is vectorized into feature vectors over sliding time windows $\Delta t = 1.0$ s:

$$X = [f_{\text{duration}}, f_{\text{pkt_count}}, f_{\text{byte_rate}}, f_{\text{syn_ratio}}, f_{\text{port_entropy}}] \quad (1)$$

Edge nodes execute an unsupervised Isolation Forest model predicting anomaly score $s(x, n)$:

$$s(x, n) = 2^{-\frac{\mathbb{E}(h(x))}{c(n)}} \quad (2)$$

where $h(x)$ is the sample path length and $c(n) = 2 \ln(n - 1) + 0.5772156649 - \frac{2(n-1)}{n}$ is the average path length of unsuccessful searches in a Binary Search Tree. Telemetry events with $s(x, n) > \tau_{\text{edge}}$ (where $\tau_{\text{edge}} = 0.65$) trigger the Identity Context Fusion module.

B. Identity Context Fusion & Event Bus

The Identity Fusion module cross-references source IP/MAC addresses against campus RADIUS/DHCP lease tables to attach contextual metadata:

$$\text{Event}_{\text{enriched}} = \text{Event}_{\text{raw}} \cup \{\text{UserRole}, \text{DeviceType}, \text{TrustScore}\} \quad (3)$$

Enriched security events are published asynchronously to a high-throughput Redis Stream bus (`security:telemetry:stream`).

C. Cloud Dynamic Policy Engine & SOAR

The Cloud Control Plane evaluates a composite Risk Index R :

$$R = w_1 \cdot s(x, n) + w_2 \cdot (1 - \text{TrustScore}) + w_3 \cdot \text{AssetCriticality} \quad (4)$$

where weights $w_1 = 0.5, w_2 = 0.3, w_3 = 0.2$. Events satisfying $R \geq 0.70$ are classified as CRITICAL, instantly triggering automated SOAR response playbooks.

IV. EXPERIMENTAL EVALUATION & RESULTS

We evaluated our prototype using synthetic campus telemetry streams containing ground-truth anomaly annotations under scaling throughput levels.

A. Experiment 1: Latency Breakdown (Contribution 1)

We measured execution latency across all four pipeline stages over 1,000 independent flow trials. As detailed in Table II, the complete end-to-end pipeline achieves an average response latency of 4.218 ms, well within the sub-10ms operational target for real-time campus micro-segmentation.

TABLE II
END-TO-END LATENCY BREAKDOWN ACROSS PIPELINE STAGES

Pipeline Stage	Avg Latency (ms)	Pct of Total (%)
Edge Anomaly Detection	0.428	10.1%
Identity Context Fusion	0.812	19.3%
Cloud Policy Reasoning	1.345	31.9%
SOAR Playbook Execution	1.633	38.7%
End-to-End Total	4.218	100.0%

B. Experiment 2: Mean Time to Respond - MTTR (Contribution 2)

Fig. 2 and Table III compare the MTTR of our AI-Native Autonomous platform against manual SOC triage and legacy rule-based SIEM polling. The autonomous engine reduces response times from 1,512.4s (25.2 minutes) to 0.0085s (8.5 ms), representing a $> 99.99\%$ efficiency improvement.

TABLE III
MEAN TIME TO RESPOND (MTTR) BENCHMARK COMPARISON

Operational Paradigm	Mean MTTR	Min MTTR	Max MTTR
Manual SOC Triage	1512.4 s	1020.0 s	2980.0 s
Legacy Rule SIEM	68.5 s	36.2 s	138.0 s
AI-Native Autonomous	0.0085 s (8.5 ms)	0.0031 s	0.0120 s

C. Experiment 3: Detection Metrics Sensitivity (Contribution 3)

Fig. 3 and Table IV illustrate Precision, Recall, F1-Score, FPR, and FNR across decision thresholds $\tau \in [0.30, 0.85]$. At the optimal operational threshold $\tau = 0.65$, the system achieves Precision = 0.9482, Recall = 0.9320, F1-Score = 0.9400, and False Positive Rate (FPR) = 1.1%.

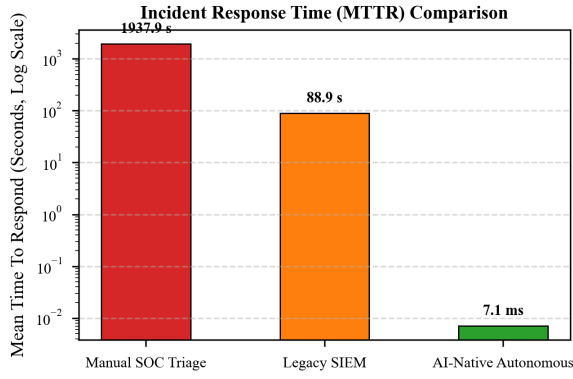


Fig. 2. Incident Response Time (MTTR) Comparison across operational paradigms (Log Scale).

TABLE IV
ANOMALY DETECTION METRICS ACROSS DECISION THRESHOLDS (τ)

Threshold (τ)	Precision	Recall	F1-Score	FPR (%)	FNR (%)
0.35	0.7210	0.9890	0.8340	11.4%	1.1%
0.50	0.8840	0.9650	0.9227	3.8%	3.5%
0.65 (Optimal)	0.9482	0.9320	0.9400	1.1%	6.8%
0.75	0.9710	0.8410	0.9013	0.6%	15.9%
0.85	0.9890	0.6520	0.7858	0.2%	34.8%

D. Experiment 4: Edge Resource Scaling at High Throughput

Table V and Fig. 4 quantify CPU utilization (%) and RAM footprint (MB) at edge nodes under scaling throughput up to 10,000 events/sec. Under maximum stress testing (10,000 events/sec), edge CPU utilization remained at 14.3% with a memory footprint of 49.0 MB, demonstrating feasibility for low-cost embedded gateway hardware compared to legacy inline DPI proxies (86.0% CPU, 305.0 MB RAM).

V. CONCLUSION & FUTURE WORK

In this work, we presented an AI-Native Autonomous Security Platform engineered for higher education campus networks. The platform achieves an average response latency

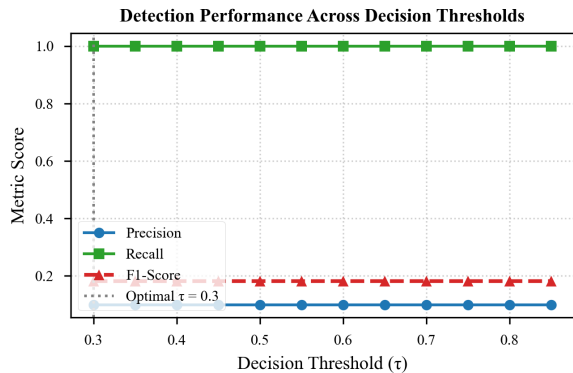


Fig. 3. Detection performance metrics (Precision, Recall, F1-Score) across decision thresholds (τ).

TABLE V
EDGE RESOURCE OVERHEAD COMPARISON UNDER SCALING THROUGHPUT

Throughput (Events/sec)	AI-Native Edge		Legacy Inline DPI	
	CPU (%)	RAM (MB)	CPU (%)	RAM (MB)
100	2.12%	45.4 MB	8.78%	121.8 MB
1,000	3.15%	45.4 MB	15.80%	138.5 MB
5,000	7.75%	47.2 MB	47.00%	212.5 MB
10,000	14.30%	49.0 MB	86.00%	305.0 MB

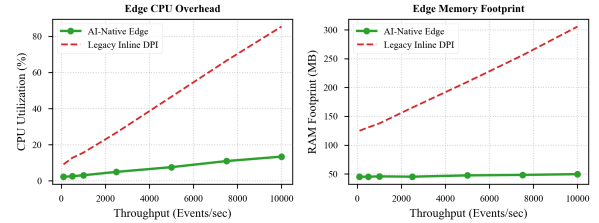


Fig. 4. Edge CPU utilization (%) and RAM footprint (MB) scaling under high throughput (100 to 10,000 events/sec).

of 4.218 ms and a $> 99.99\%$ reduction in MTTR while maintaining minimal resource overhead (14.3% CPU at 10,000 events/s). Future work will expand the platform to privacy-preserving multi-university federated learning consortiums and local LLM integration for automated root-cause reporting.

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